

Deep Exposition: Fourier–Bessel Functions of Singular Continuous Measures and Their Many Asymptotics

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1. Statement of the Subject

The paper defines a family of special functions—called Fourier–Bessel (F-B) functions—associated to an arbitrary positive Borel measure μ for which the Hamburger moment problem is determinate. Given the orthonormal polynomial sequence $\{p_n(\mu; s)\}_{n \in \mathbb{N}}$ determined by μ , the n -th Fourier–Bessel function is

$$J_n(\mu; t) := \int p_n(\mu; s) e^{-its} d\mu(s). \quad (1.1)$$

The paper investigates the asymptotic behavior of these functions in three distinct regimes:

1. Large argument $t \rightarrow \infty$ at fixed order n : For singular continuous measures, the functions $J_n(\mu; t)$ do not decay to zero but exhibit "quantum intermittency"—bursts of large amplitude interspersed with intervals of near-zero values. After Cesàro (or Gaussian) time-averaging, algebraic decay emerges, governed by dimensional quantities of μ .
2. Large order $n \rightarrow \infty$ at fixed argument t : The functions decay super-exponentially:

$$|J_n(\mu; t)| \leq C_\alpha e^{-\alpha n} \quad \text{for all } n, \quad (1.2)$$

for every $\alpha \geq 0$, provided $\text{supp}(\mu)$ is bounded.

3. Simultaneous asymptotics (large n and large t in linked fashion): The growth of position moments

$$\nu_\alpha(t) := \sum_{n=0}^{\infty} n^\alpha |J_n(\mu; t)|^2, \quad (1.8)$$

quantified by the quantum intermittency exponents $\beta^\pm(\alpha)$, is related to the Rényi generalized dimension spectrum $D_q(\mu)$. For Julia set measures of the quadratic family $z \mapsto z^2 - \lambda$ (with $\lambda > 2$), a renormalization argument yields the fundamental Julia set relation:

$$\beta(\alpha) = D_{1-\alpha}(\mu). \quad (11.7)$$

The paper presents established results, new derivations via renormalization of the iterated function system (IFS) generating the Julia set, and two conjectures extending the relationship between $\beta(\alpha)$ and $D_q(\mu)$ to broader classes of IFS measures.

2. Prerequisites

2.1 Orthogonal Polynomials and the Moment Problem

Let μ be a positive Borel probability measure on $\mathbb{R}$ with finite moments of all orders: $\int |s|^k d\mu(s) < \infty$ for every $k \in \mathbb{N}$. The Hamburger moment problem is determinate for μ if μ is the unique measure having the moment sequence $\{m_k\}_{k=0}^\infty$, where $m_k = \int s^k d\mu(s)$. Determinacy is guaranteed whenever $\text{supp}(\mu)$ is compact (Weierstrass approximation).

Gram–Schmidt orthonormalization of the sequence $1, s, s^2, \dots$ in $L^2(\mu)$ produces the orthonormal polynomial sequence $\{p_n(\mu; s)\}_{n=0}^\infty$ satisfying

$$\int p_n(\mu; s) p_m(\mu; s) d\mu(s) = \delta_{nm}.$$

2.2 Three-Term Recurrence and the Jacobi Matrix

Every orthonormal polynomial sequence satisfies a three-term recurrence:

$$s p_n(\mu; s) = a_{n+1} p_{n+1}(\mu; s) + b_n p_n(\mu; s) + a_n p_{n-1}(\mu; s), \quad (\text{R})$$

with $a_n > 0$ and $b_n \in \mathbb{R}$. In vector notation, writing $\mathbf{p}(\mu; s) = (p_0(\mu; s), p_1(\mu; s), \dots)^T$,

$$s \mathbf{p}(\mu; s) = J_\mu \mathbf{p}(\mu; s), \quad (1.3)$$

where J_μ is the Jacobi matrix (tridiagonal, symmetric):

$$J_\mu = \begin{pmatrix} b_0 & a_1 & 0 & \cdots \\ a_1 & b_1 & a_2 & \cdots \\ 0 & a_2 & b_2 & \cdots \\ \vdots & & & \ddots \end{pmatrix}.$$

By the spectral theorem, J_μ is a bounded self-adjoint operator on $\ell^2(\mathbb{Z}_+)$ when $\text{supp}(\mu)$ is bounded, and its spectral measure with respect to the canonical basis vector $e_0 = (1, 0, 0, \dots)^T$ is precisely μ .

2.3 Schrödinger Equation on the Lattice

The Schrödinger equation

$$i \frac{d}{dt} \psi(t) = J_\mu \psi(t), \quad (1.4)$$

with initial condition $\psi(0) = e_0$, has solution $\psi(t) = e^{-iJ_\mu t} e_0$. The projection onto the n -th basis state is

$$\psi_n(t) = \langle e_n, e^{-iJ_\mu t} e_0 \rangle = \int p_n(\mu; s) e^{-its} d\mu(s) = J_n(\mu; t). \quad (1.6)$$

This identity is the physical interpretation of F-B functions: $J_n(\mu; t)$ is the quantum-mechanical amplitude for a particle, initially localized at site 0 of a one-dimensional lattice with Hamiltonian J_μ , to be found at site n at time t .

2.4 Unitarity and Probability Conservation

Since $e^{-iJ_\mu t}$ is unitary,

$$\sum_{n=0}^{\infty} |J_n(\mu; t)|^2 = \|e^{-iJ_\mu t} e_0\|^2 = \|e_0\|^2 = 1. \quad (1.7)$$

The quantities $|J_n(\mu; t)|^2$ define a probability distribution on $\mathbb{Z}_+$ for each t .

2.5 Position Moments and Growth Exponents

The α -th moment of the position in this probability distribution is

$$\nu_\alpha(t) := \sum_{n=0}^{\infty} n^\alpha |J_n(\mu; t)|^2. \quad (1.8)$$

The **growth exponents** (or quantum dynamical exponents) are:

$$\beta^\pm(\alpha) = \frac{1}{\alpha} \limsup_{t \rightarrow \infty} \frac{\log \nu_\alpha(t)}{\log t}. \quad (1.9)$$

(resp. $\liminf$)

These measure the power-law rate at which the quantum wave-packet spreads. By unitarity (eq. 1.7), $\nu_\alpha(t) \leq (\sup_n n : |J_n(\mu; t)| > 0)^\alpha$, which, combined with the finite propagation speed for bounded Jacobi matrices, yields $\beta^\pm(\alpha) \leq 1$. The value $\beta = 1$ corresponds to ballistic transport.

2.6 Chebyshev Polynomials and Classical Bessel Functions

The Chebyshev measure on $[-1, 1]$ is $d\mu(s) = \frac{ds}{\pi\sqrt{1-s^2}}$. Its orthonormal polynomials are proportional to the Chebyshev polynomials of the first kind T_n . The corresponding F-B functions reduce to classical Bessel functions:

$$J_n(\mu; t) = (-i)^n J_n(t),$$

where $J_n(t)$ is the Bessel function of the first kind of integer order n . The well-known identity $\sum_{n=-\infty}^{\infty} J_n(t)^2 = 1$ (Neumann's addition theorem) is the specialization of eq. (1.7). Ballistic spreading $\beta^\pm(\alpha) = 1$ holds for this absolutely continuous measure.

2.7 Generalized (Rényi) Dimensions

For a Borel probability measure μ and $q > 0, q \neq 1$, the generalized Rényi dimension of order q is defined via the scaling of local measure integrals. Let $B_\omega(r)$ denote the ball (interval) of radius ω centered at r . Define:

$$I_q(\mu; \omega) := \int (\mu(B_\omega(r)))^{q-1} d\mu(r).$$

Then

$$D_q(\mu) := \frac{1}{q-1} \lim_{\omega \rightarrow 0} \frac{\log I_q(\mu; \omega)}{\log \omega}, \quad (5.1)$$

provided the limit exists (otherwise one takes $\limsup$ or $\liminf$ to get upper and lower dimensions $\overline{D}_q, \underline{D}_q$).

Key properties:

- $D_q(\mu)$ is non-increasing in q .
- $D_0(\mu)$ is the box-counting (Minkowski) dimension of $\text{supp}(\mu)$.
- $D_1(\mu) = \lim_{q \rightarrow 1} D_q(\mu)$ is the information dimension (entropy dimension).
- $D_2(\mu)$ is the correlation dimension.
- For an absolutely continuous measure with bounded density on a set of dimension d , $D_q = d$ for all q .

Alternative characterization via coverings: Cover $\text{supp}(\mu)$ by disjoint intervals $\{I_\sigma\}$ of lengths $\{l_\sigma\}$ and masses $\pi_\sigma := \mu(I_\sigma)$. Define the generalized energy function

$$H(x, \Sigma) := \sum_{\sigma \in \Sigma} \pi_\sigma^q l_\sigma^{(1-q)x}. \quad (5.2)$$

Then $D_q(\mu)$ is the divergence abscissa of this function: the critical value of x at which $H(x, \Sigma)$ transitions from convergent to divergent as the covering is refined.

2.8 Hausdorff and Packing Dimensions

The Hausdorff dimension $\dim_H(\mu)$ and packing dimension $\dim_p(\mu)$ of a measure μ are defined as:

$$\dim_H(\mu) = \inf\{\dim_H(A) : \mu(A) > 0\}, \quad \dim_p(\mu) = \inf\{\dim_p(A) : \mu(\mathbb{R} \setminus A) = 0\},$$

where $\dim_H(A)$ and $\dim_p(A)$ are the Hausdorff and packing dimensions of the set A . One has the chain of inequalities:

$$D_2(\mu) \leq D_1(\mu) \leq \dim_H(\mu) \leq \dim_p(\mu) \leq D_0(\mu).$$

2.9 Gaussian Averaging

Define the Gaussian time-average of a function f :

$$A_G(f)(t) := \frac{1}{2t\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2/t^2} f(s) ds. \quad (2.3)$$

This is a smooth substitute for the Cesàro average $\frac{1}{T} \int_0^T f(s) ds$. Introduce the Gaussian kernel:

$$\chi_\omega(u) := e^{-u^2/\omega^2},$$

which is a smooth analogue of the characteristic function of $[-\omega, \omega]$. The convention throughout is $\omega := t^{-1}$, so large time corresponds to small ω .

2.10 Iterated Function Systems (IFS) and Balanced Measures

An iterated function system (IFS) consists of a finite collection of contracting maps $\{l_i\}_{i=1}^M$ on $\mathbb{R}$ with associated probabilities $\{\pi_i\}_{i=1}^M$, $\pi_i > 0$, $\sum \pi_i = 1$. The unique invariant (balanced) measure μ satisfies the balance equation:

$$\int f(s) d\mu(s) = \sum_{i=1}^M \pi_i \int (f \circ l_i)(s) d\mu(s). \quad (12.1)$$

For a linear IFS with $l_i(s) = \delta_i s + \theta_i$, $0 < |\delta_i| < 1$, the support of μ is the attractor of the IFS—a self-similar set, which is a Cantor set when the images $l_i(\text{supp}(\mu))$ are disjoint (the open set condition holds).

2.11 Julia Sets and the Quadratic Family

For $\lambda \geq 2$, the real Julia set of $z \mapsto z^2 - \lambda$ is the maximal bounded invariant set under iteration. It is the attractor of the IFS with two inverse branches:

$$\varphi_j(s) = j\sqrt{s + \lambda}, \quad j = \pm 1. \quad (9.1)$$

The balanced measure (also called the measure of maximal entropy, or the Brolin measure restricted to the real line) satisfies:

$$\int f(x) d\mu(x) = \frac{1}{2} \sum_{j=\pm 1} \int (f \circ \varphi_j)(x) d\mu(x). \quad (9.2)$$

At $\lambda = 2$, the Julia set is $[-2, 2]$ and μ is the (rescaled) Chebyshev measure, so it is absolutely continuous. For $\lambda > 2$, the Julia set is a Cantor set and μ is singular continuous.

The orthogonal polynomials of the Julia set measure satisfy a polynomial renormalization property:

$$p_{2n}(\mu; \varphi_j(s)) = p_n(\mu; s). \quad (9.4)$$

This is a consequence of the algebraic structure of the Julia set and the relation between the Chebyshev polynomials $T_{2n}(x) = T_n(2x^2 - 1)$ at $\lambda = 2$, extended by continuity in λ .

2.12 Reproducing Kernel

The reproducing kernel (Christoffel–Darboux kernel) is

$$K_\mu^0(r, s) := \sum_{n=0}^{\infty} p_n(\mu; s) p_n(\mu; r).$$

By orthonormality, this acts as a reproducing kernel: for any polynomial f of degree $\leq N$,

$$\int K_{\mu, N}^0(r, s) f(s) d\mu(s) = f(r),$$

where $K_{\mu, N}^0(r, s) = \sum_{n=0}^N p_n(\mu; s) p_n(\mu; r)$. In the distributional sense for the full sum, $K_\mu^0(r, s) = \delta_\mu(r - s)$, the Dirac delta with respect to μ .

2.13 Guarneri's Inequality and Refinements

Guarneri's inequality (1993): For the Schrödinger dynamics on a Jacobi matrix,

$$\beta^-(\alpha) \geq \dim_H(\mu)$$

for all $\alpha > 0$. This was refined by Guarneri–Schulz–Baldes and by Barbaroux–Germinet–Tcheremchantsev to:

$$\beta^-(\alpha) \geq D_{(1+\alpha)^{-1}}(\mu). \tag{3.2}$$

Since $D_q(\mu)$ is non-increasing in q and $(1 + \alpha)^{-1} < 1$ for $\alpha > 0$, one has $D_{(1+\alpha)^{-1}} \geq D_1 \geq \dim_H(\mu)$, so eq. (3.2) strengthens Guarneri's original bound.

2.14 Logarithmic Potential and Equilibrium Measure

The logarithmic potential of a measure ν supported on a compact set $K \subset \mathbb{R}$ is

$$U^\nu(z) = - \int \log |z - s| d\nu(s).$$

The equilibrium measure ν_K minimizes the energy $\iint (-\log |r - s|) d\nu(r) d\nu(s)$ among all probability measures supported on K . For Cantor sets arising from IFS, the equilibrium measure encodes the geometric distribution of the set, independent of any probability weights π_i .

3. Complete Exegesis

The argument of the paper proceeds through several stages, each establishing asymptotic properties of the F-B functions in a different regime. The exposition below follows the paper's logical arc.

3.1 Definition and Classical Reduction (Section 1)

Step 1. The F-B functions are defined by eq. (1.1). When μ is the Chebyshev measure $d\mu(s) = \frac{ds}{\pi\sqrt{1-s^2}}$ on $[-1, 1]$, the orthonormal polynomials are $p_n(\mu; s) = \sqrt{2} T_n(s)$ for $n \geq 1$ (with $p_0 = 1$), and one computes:

$$J_n(\mu; t) = \int_{-1}^1 p_n(\mu; s) e^{-its} \frac{ds}{\pi\sqrt{1-s^2}} = (-i)^n J_n(t),$$

by the standard integral representation of integer-order Bessel functions. The factor $(-i)^n$ arises from the phase convention in the Fourier kernel e^{-its} combined with the parity of Chebyshev polynomials: $T_n(\cos \theta) = \cos(n\theta)$, and the integral reduces to the Bessel integral $J_n(t) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - t \cos \theta) d\theta$.

Step 2. For a symmetric measure (i.e., μ invariant under $s \mapsto -s$), $p_n(\mu; s)$ has the same parity as n . Splitting the Fourier kernel into real and imaginary parts, $e^{-its} = \cos(ts) - i \sin(ts)$, and using parity of the integrand: for n even, only the cosine part survives (giving a real J_n); for n odd, only the sine part survives (giving a purely imaginary J_n).

3.2 Connection to Quantum Dynamics (Section 1)

Step 3. The Jacobi matrix J_μ generates quantum dynamics via eq. (1.4). The solution is $\psi(t) = e^{-iJ_\mu t} e_0$. Expanding in the eigenbasis of J_μ —which, by the spectral theorem, is identified with the multiplication operator $(Mf)(s) = sf(s)$ on $L^2(\mu)$ via the unitary map $e_n \mapsto p_n(\mu; \cdot)$ —the n -th component is:

$$\psi_n(t) = \langle e_n, e^{-iJ_\mu t} e_0 \rangle = \int p_n(\mu; s) \overline{p_0(\mu; s)} e^{-its} d\mu(s).$$

Since $p_0(\mu; s) = 1$ (constant, by normalization $\int d\mu = 1$), this yields $\psi_n(t) = J_n(\mu; t)$. The step from the abstract spectral theorem to this integral uses: (i) the spectral measure of J_μ with respect to e_0 is μ ; (ii) the functional calculus gives $\langle e_n, f(J_\mu) e_0 \rangle = \int p_n(\mu; s) f(s) d\mu(s)$ for any bounded Borel function f .

Step 4. Unitarity of $e^{-iJ_\mu t}$ immediately implies eq. (1.7): $\sum_n |J_n(\mu; t)|^2 = 1$. This is a consequence of Parseval's identity: the map $e_n \mapsto p_n$ is unitary from ℓ^2 to $L^2(\mu)$, and e^{-its} has unit $L^2(\mu)$ -norm (since μ is a probability measure).

3.3 Super-Exponential Decay in Order (Section 1)

Step 5. When $\text{supp}(\mu) \subseteq [-R, R]$ is bounded, the orthogonal polynomials $p_n(\mu; s)$ grow at most exponentially in n on $\text{supp}(\mu)$. More precisely, since $\|p_n\|_{L^2(\mu)} = 1$ and p_n is a polynomial of exact degree n , while the Fourier kernel e^{-its} is entire in s , the integral $J_n(\mu; t) = \int p_n(\mu; s) e^{-its} d\mu(s)$ is estimated using the Cauchy–Schwarz inequality:

$$|J_n(\mu; t)| = \left| \int p_n(\mu; s) e^{-its} d\mu(s) \right| \leq \|p_n\|_{L^2(\mu)} \|e^{-it\cdot}\|_{L^2(\mu)} = 1.$$

This gives only a uniform bound. The super-exponential decay $|J_n(\mu; t)| \leq C_\alpha e^{-\alpha n}$ for every $\alpha \geq 0$ (eq. 1.2) requires a deeper argument. It follows from the fact that for compactly supported measures, the leading coefficient of p_n grows like $\text{cap}(\text{supp}(\mu))^{-n}$ (where cap is the logarithmic capacity), and the oscillatory integral against the smooth kernel e^{-its} produces cancellation that overwhelms any exponential growth. The key ingredient is that e^{-its} is an entire function of s , so it lies in the domain of arbitrarily high powers of J_μ , and iterated integration by parts (in the sense of the recurrence relation) transfers powers of n into inverse powers of t with controllable remainders. The decay is faster than any exponential in n at fixed t .

3.4 Kinematics: Expansion in Orthogonal Polynomials (Section 2)

Step 6. Inserting the definition $J_n(\mu; t) = \int p_n(\mu; s) e^{-its} d\mu(s)$ into the position moments:

$$\nu_\alpha(t) = \sum_{n=0}^{\infty} n^\alpha |J_n(\mu; t)|^2 = \sum_{n=0}^{\infty} n^\alpha \int \int e^{i(r-s)t} p_n(\mu; s) p_n(\mu; r) d\mu(s) d\mu(r). \quad (1)$$

The interchange of sum and double integral is justified by absolute convergence: the super-exponential decay of $|J_n|$ in n (eq. 1.2) ensures $\sum n^\alpha |J_n|^2 < \infty$ for all α . The double integral represents $|J_n|^2$ as a bilinear form in the product measure $\mu \otimes \mu$.

Step 7. Define the weighted kernel:

$$K_\mu^\alpha(r, s) := \sum_{n=0}^{\infty} n^\alpha p_n(\mu; s) p_n(\mu; r). \quad (2.2)$$

For $\alpha = 0$, this is the reproducing kernel $K_\mu^0(r, s) = \delta_\mu(r - s)$. Then:

$$\nu_\alpha(t) = \int \int e^{i(r-s)t} K_\mu^\alpha(r, s) d\mu(s) d\mu(r).$$

Step 8. Gaussian time-averaging (eq. 2.3) replaces the oscillatory factor $e^{i(r-s)t}$ with the Gaussian kernel $\chi_{1/t}(r - s) = e^{-(r-s)^2 t^2}$ (after computing the Gaussian transform of an exponential). Explicitly:

$$A_G(\nu_\alpha)(t) = \sum_{n=0}^{\infty} n^\alpha \int \int \chi_\omega(r - s) p_n(\mu; s) p_n(\mu; r) d\mu(s) d\mu(r), \quad (2.4)$$

where $\omega = t^{-1}$. The derivation uses: the Gaussian average of $e^{i(r-s)t}$ in t , with the Gaussian weight $\frac{1}{2t\sqrt{\pi}} e^{-s^2/t^2}$, yields $e^{-(r-s)^2/(4\omega^2)}$ up to normalization constants (by completing the square in the exponent). The precise form $\chi_\omega(r-s) = e^{-(r-s)^2/\omega^2}$ depends on the normalization convention chosen for A_G .

3.5 Distribution Functions and Lower Bounds (Section 3)

Step 9. Define the truncated zeroth moment:

$$\nu_0(N, \omega) := \int \int \chi_\omega(r-s) \sum_{n=0}^N p_n(\mu; s) p_n(\mu; r) d\mu(s) d\mu(r). \quad (3.1)$$

This equals the Gaussian time average (up to $t = \omega^{-1}$) of $\sum_{n=0}^N |J_n(\mu; t)|^2$. Since $\sum_{n=0}^\infty |J_n|^2 = 1$, one has $\nu_0(N, \omega) \leq 1$ and $\nu_0(\infty, \omega) = 1$.

Step 10. Guarneri's method proceeds by contrapositive: if the wave-packet is too concentrated on $\{0, 1, \dots, N\}$, then $\nu_0(N, \omega)$ is close to 1, and bounding ν_0 away from 1 forces the wave to spread. Concretely, $1 - \nu_0(N, \omega)$ is a lower bound for the probability of being beyond site N . Upper bounds on $\nu_0(N, \omega)$ as a function of N and ω translate, via optimization over N , into lower bounds on $\beta^-(\alpha)$. The refinement eq. (3.2) uses the generalized dimension $D_{(1+\alpha)^{-1}}(\mu)$, exploiting the fact that higher moments are more sensitive to the tail of the distribution, and thus to finer dimensional properties of μ .

3.6 Ketzmerick et al. Scaling and Further Lower Bounds (Section 4)

Step 11. Ketzmerick, Kruse, Kraut, and Geisel proposed the scaling ansatz:

$$\nu_0(N, \omega) \sim N^d \omega^{D_2(\mu)}, \quad (4.1)$$

valid in the intermediate regime $1 \ll N \ll \omega^{-D_2(\mu)/d}$. The exponent d encodes growth properties of orthogonal polynomials.

Step 12. If a uniform bound

$$\nu_0(N, \omega) \leq c N^\gamma \omega^{D_2(\mu)} \quad (4.2)$$

holds, then optimizing: the probability beyond site N is at least $1 - c N^\gamma \omega^{D_2(\mu)}$. For ν_α , the contribution from sites $n > N$ is at least $N^\alpha (1 - c N^\gamma \omega^{D_2})$. Choosing N to maximize the lower bound (i.e., $N \sim \omega^{-D_2/\gamma}$) gives:

$$\nu_\alpha(t) \gtrsim t^{\alpha D_2(\mu)/\gamma},$$

hence $\beta^-(\alpha) \geq D_2(\mu)/\gamma$ (eq. 4.3).

3.7 Correlation Dimension and Averaged Squared F-B Functions (Section 5)

Step 13. Starting from the Gaussian-averaged square of a single F-B function:

$$A_G(|J_n(\mu; t)|^2) = \int \int \chi_\omega(r - s) p_n(\mu; s) p_n(\mu; r) d\mu(s) d\mu(r). \quad (5.3)$$

When $\omega \ll 1/n$ —meaning the Gaussian kernel $\chi_\omega(r - s)$ concentrates on $|r - s| \lesssim \omega$ while $p_n(\mu; \cdot)$ varies on a scale $\sim 1/n$ —the polynomial is approximately constant over the ball $B_\omega(r)$, so $p_n(\mu; s) \approx p_n(\mu; r)$ for $|r - s| \leq \omega$. This yields:

$$A_G(|J_n(\mu; t)|^2) \approx \int p_n^2(\mu; r) \mu(B_\omega(r)) d\mu(r). \quad (5.4)$$

Justification of the approximation: The polynomial p_n is Lipschitz on $\text{supp}(\mu)$ with Lipschitz constant bounded by Cn^2 (Bernstein–Markov inequality for compactly supported measures). The Gaussian kernel χ_ω restricts the effective integration to $|r - s| \lesssim \omega$. The error in replacing $p_n(\mu; s)$ by $p_n(\mu; r)$ is $O(n^2\omega)$, which is negligible when $\omega \ll 1/n$.

Step 14. The integral $\int p_n^2(\mu; r) \mu(B_\omega(r)) d\mu(r) = \int (\mu(B_\omega(r)))^{2-1} d\rho_n(r)$, where $d\rho_n(r) := p_n^2(\mu; r) d\mu(r)$ is a probability measure (by orthonormality). The scaling $\int (\mu(B_\omega(r)))^{q-1} d\mu(r) \sim \omega^{(q-1)D_q(\mu)}$ (eq. 5.1) with $q = 2$ gives the decay rate $\omega^{D_2(\mu)}$, provided the measure ρ_n does not concentrate on exceptional sets where the local dimension of μ deviates from D_2 . This connects the correlation dimension directly to the decay of averaged $|J_n|^2$.

3.8 Asymptotics of Orthogonal Polynomials and Growth Exponents (Section 6)

Step 15. Applying the same small- ω approximation to the truncated moment:

$$\nu_0(N, \omega) \approx \int \mu(B_\omega(r)) \sum_{n=0}^N p_n^2(\mu; r) d\mu(r). \quad (6.1)$$

This factorization separates the problem into two parts: (a) the growth of the partial sums $\sum_{n=0}^N p_n^2(\mu; r)$ as $N \rightarrow \infty$, and (b) the scaling of $\mu(B_\omega(r))$ as $\omega \rightarrow 0$.

Step 16. If the partial sums grow algebraically with a uniform exponent:

$$\sum_{n=0}^N p_n^2(\mu; r) \leq C N^\gamma \quad \text{for all } r \in \text{supp}(\mu), \quad (6.4)$$

then:

$$\nu_0(N, \omega) \leq C N^\gamma \int \mu(B_\omega(r)) d\mu(r) = C N^\gamma I_2(\mu; \omega) \sim C N^\gamma \omega^{D_2(\mu)}.$$

Inserting this into the optimization argument of Step 12 with general α : the α -th moment receives contributions from sites $n > N$ of order $N^\alpha(1 - CN^\gamma\omega^{D_2})$. Maximizing over N at fixed $\omega = t^{-1}$ yields:

$$\beta^-(\alpha) \geq \frac{1}{\gamma} D_{1-\alpha/\gamma}(\mu). \quad (6.5)$$

Derivation of (6.5): The full optimization uses the generalized dimension spectrum, not just D_2 . The integral $\int \mu(B_\omega(r)) \sum_{n=0}^N p_n^2(\mu; r) d\mu(r)$ is bounded by $CN^\gamma \int \mu(B_\omega(r)) d\mu(r)$. When one needs a sharper bound involving D_q for $q \neq 2$, the Hölder inequality is applied to the integrand $\mu(B_\omega(r))^{q-1}$ with appropriate exponents, leading to the q -dependent family of bounds that, after optimization over q , yield eq. (6.5) with the specific index $q = 1 - \alpha/\gamma$. The constraint $q > 0$ requires $\alpha < \gamma$.

Step 17. The local version (eq. 6.6): if a Borel set $S \subseteq \text{supp}(\mu)$ exists with $\mu(S) > 0$, the restricted measure $\mu|_S$ is a -continuous (meaning $\mu(B_\omega(r)) \leq C\omega^a$ for $r \in S$), and eq. (6.4) holds for $r \in S$, then $\beta^-(\alpha) \geq a/\gamma$. This follows because the integral in eq. (6.1), restricted to S , is bounded by $CN^\gamma \omega^a \mu(S)$, giving a lower bound on the spreading rate determined by the local regularity exponent a rather than the global dimension D_2 .

3.9 Upper Bounds from Sparse Barriers (Section 7)

Step 18. Lower bounds on $\nu_0(N, \omega)$ do not produce upper bounds on $\beta(\alpha)$, because a large value of ν_0 only means the wave-packet remains concentrated—it does not prevent spreading. Upper bounds on β require a different mechanism.

Step 19. For sparse barrier Jacobi matrices, defined by:

$$x p_k(x; \mu) = (V_\eta(k) + \theta \delta_{0,k}) p_k(x; \mu) + p_{k-1}(x; \mu) + p_{k+1}(x; \mu), \quad (7.1)$$

with potential $V_\eta(k) = \chi_B(k) k^\eta$ (barriers of height k^η at a sparse set $B = \{L_n\}$ with $L_{n+1} \geq aL_n, a > 1$), one obtains:

$$\beta^-(\alpha) \leq \frac{\alpha + 1}{2\eta + \alpha + 1}, \quad (7.2)$$

while $\beta^+(\alpha) = 1$. The barriers create partial reflections that slow the wave-packet on average (reducing β^-) but allow occasional ballistic excursions (preserving $\beta^+ = 1$). This separation between β^- and β^+ is the hallmark of quantum intermittency.

3.10 Wave-Front Propagation (Section 8)

Step 20. Numerical evidence indicates that the outermost occupied site (the "wave front") propagates as $n_{\text{front}} \sim t^\eta$ with $\eta < 1$ for singular continuous measures. In the numerical example given, $\eta = 0.84165$. This provides the upper bound $\beta^+(\alpha) \leq \eta$. Determining η analytically from the properties of μ remains an open problem.

3.11 Renormalization for Julia Set Measures (Section 9)

Step 21. For Julia set measures of $z \mapsto z^2 - \lambda$, the polynomial renormalization (eq. 9.4) and the IFS balance equation (eq. 9.2) are combined. Define the Gaussian-averaged wave projection:

$$\psi_n^G(t) := A_G(|J_n(\mu; t)|^2).$$

Using eq. (9.4), $p_{2n}(\mu; \varphi_j(s)) = p_n(\mu; s)$, and applying eq. (9.2) twice (once for each variable in the double integral of eq. 5.3):

$$\psi_{2n}^G(t) = \frac{1}{4} \sum_{\sigma, \sigma' = \pm 1} \int \int \chi_\omega(\varphi_\sigma(r) - \varphi_{\sigma'}(s)) p_n(\mu; r) p_n(\mu; s) d\mu(s) d\mu(r). \quad (9.5)$$

The factor $1/4 = (1/2)^2$ arises from applying the balance equation with equal weights $\pi_j = 1/2$ in each integration variable independently.

Step 22. After k iterations, at site $N = n \cdot 2^k$:

$$\psi_N^G(t) = \frac{1}{2^{2k}} \sum_{\substack{\sigma, \sigma' \\ |\sigma| = |\sigma'| = k}} \int \int \chi_\omega(\varphi_\sigma(r) - \varphi_{\sigma'}(s)) p_n(\mu; r) p_n(\mu; s) d\mu(s) d\mu(r), \quad (9.6)$$

where $\sigma = (\sigma_1, \dots, \sigma_k)$ is a multi-index with $\sigma_i \in \{+1, -1\}$, and $\varphi_\sigma = \varphi_{\sigma_1} \circ \dots \circ \varphi_{\sigma_k}$. The factor $1/2^{2k}$ accumulates from k applications of the balance equation in each variable.

Step 23. The diagonal approximation: when $\sigma \neq \sigma'$, the images $\varphi_\sigma(\text{supp}(\mu))$ and $\varphi_{\sigma'}(\text{supp}(\mu))$ are separated by a distance proportional to the gap structure of the Cantor set. For sufficiently small ω (large t), the Gaussian kernel $\chi_\omega(\varphi_\sigma(r) - \varphi_{\sigma'}(s))$ is exponentially small when $\sigma \neq \sigma'$, because $|\varphi_\sigma(r) - \varphi_{\sigma'}(s)|$ is bounded below by the gap size, which exceeds ω . Retaining only $\sigma = \sigma'$ terms:

Step 24. Linearization: Each composition map φ_σ is approximately affine on $\text{supp}(\mu)$:

$$\varphi_\sigma(s) \approx l_\sigma(s) := \delta_\sigma s + \theta_\sigma,$$

where $\delta_\sigma = \varphi'_\sigma(\text{fixed point})$ is the contraction ratio and $\theta_\sigma = \varphi_\sigma(\text{fixed point})$. Under this linearization:

$$\chi_\omega(\varphi_\sigma(r) - \varphi_\sigma(s)) \approx \chi_\omega(\delta_\sigma(r - s)) = \chi_{\omega/\delta_\sigma}(r - s).$$

The Gaussian kernel with argument $\delta_\sigma(r - s)$ and width ω is the same as a Gaussian kernel with argument $r - s$ and width ω/δ_σ . This yields the renormalization equation:

$$\psi_N^G(t) \approx \frac{1}{2^{2k}} \sum_{|\sigma|=k} \psi_n^G(t \delta_\sigma) + E(k, n, t), \quad (9.8)$$

where $E(k, n, t)$ collects the off-diagonal and linearization errors.

3.12 Behind the Wave Front: Determining γ (Section 10)

Step 25. Summing eq. (9.8) over n from 0 to some cutoff, the truncated moment satisfies:

$$\nu_0(N, \omega) \sim \sum_{|\sigma|=k} \frac{1}{2^k} \nu_0(n, \omega/\delta_\sigma). \quad (10.1)$$

Here $N = n \cdot 2^k$, and the factor $1/2^k$ (not $1/2^{2k}$) arises because one factor of 2^k is absorbed by the sum over n (the number of terms in the truncated sum grows by 2^k at each renormalization step).

Step 26. Substituting the power-law ansatz $\nu_0(N, \omega) \sim N^\gamma \omega^{D_2(\mu)}$:

$$(n \cdot 2^k)^\gamma \omega^{D_2} \sim \sum_{|\sigma|=k} \frac{1}{2^k} n^\gamma (\omega/\delta_\sigma)^{D_2}.$$

Dividing both sides by $n^\gamma \omega^{D_2}$:

$$2^{k\gamma} \sim \sum_{|\sigma|=k} \frac{1}{2^k} \delta_\sigma^{-D_2}.$$

With $\pi_\sigma = 2^{-k}$ (each multi-index of length k carries probability 2^{-k}):

$$\sum_{|\sigma|=k} \pi_\sigma^{1+\gamma} \delta_\sigma^{-D_2(\mu)} \sim 1. \quad (10.2)$$

Step 27. Comparison with the generalized energy function (eq. 5.2): setting $q = 1 + \gamma$ and requiring $H(D_q, \Sigma) \sim 1$ (the definition of D_q as the divergence abscissa):

$$\sum_{\sigma} \pi_\sigma^{1+\gamma} l_\sigma^{-\gamma D_{1+\gamma}} \sim 1.$$

Matching with eq. (10.2) requires $D_2(\mu) = \gamma D_{1+\gamma}(\mu)$. For the Julia set balanced measure with $\pi_\sigma = 2^{-k}$ and $l_\sigma = \delta_\sigma$ (uniform weights), the generalized dimensions satisfy $D_q = D_2$ for $q \geq 1$ (this is specific to the balanced measure with equal probabilities), which forces $\gamma = 1$ (eqs. 10.3–10.4). Substituting $\gamma = 1$ into eq. (4.3): $\beta^-(\alpha) \geq D_2(\mu)/1 = D_2(\mu)$. This bound is weaker than Guarneri's $\beta^- \geq D_1(\mu) \geq D_2(\mu)$, confirming that the "behind-the-wave-front" analysis alone is insufficient to capture the full spreading rate.

3.13 Surfing the Wave: The Fundamental Relation (Section 11)

Step 28. For $\alpha > 0$, define the truncated α -moment:

$$\nu_\alpha(N, \omega) := \sum_{n=0}^N n^\alpha A_G(|J_n(\mu; t)|^2). \quad (11.1)$$

Two regimes:

- **Behind the wave front** ($N \gg t^{\beta(\alpha)}$): all significant contributions are included, so $\nu_\alpha(N, \omega) \sim N^{\gamma+\alpha} \omega^{D_2(\mu)}$ (eq. 11.2), extending the $\alpha = 0$ result by the additional factor N^α from the weight n^α .
- **Ahead of the wave front** ($N \ll t^{\beta(\alpha)}$): the truncation captures only sites that the wave has not yet reached significantly, so $\nu_\alpha(N, \omega)$ is controlled by the full untruncated moment: $\nu_\alpha(N, \omega) \sim \nu_\alpha(t) \sim t^{\alpha\beta(\alpha)} = \omega^{-\alpha\beta(\alpha)}$, independent of N (eq. 11.3).

Step 29. The renormalization of the full (untruncated) α -moment proceeds differently from the $\alpha = 0$ case. At the k -th renormalization step, the site index transforms as $n \rightarrow N = n \cdot 2^k$, so the weight $N^\alpha = n^\alpha \cdot 2^{k\alpha}$ introduces an additional factor:

$$\nu_\alpha(t) \sim 2^{k(\alpha-1)} \sum_{|\sigma|=k} \nu_\alpha(\delta_\sigma t). \quad (11.5)$$

The factor $2^{k(\alpha-1)}$ combines: $2^{k\alpha}$ from the weight $N^\alpha = (n \cdot 2^k)^\alpha$ and 2^{-k} from the IFS probabilities.

Step 30. Substituting the scaling ansatz $\nu_\alpha(t) \sim t^{\alpha\beta(\alpha)}$:

$$t^{\alpha\beta(\alpha)} \sim 2^{k(\alpha-1)} \sum_{|\sigma|=k} (\delta_\sigma t)^{\alpha\beta(\alpha)}.$$

Dividing by $t^{\alpha\beta(\alpha)}$:

$$1 \sim 2^{k(\alpha-1)} \sum_{|\sigma|=k} \delta_\sigma^{\alpha\beta(\alpha)}.$$

With $\pi_\sigma = 2^{-k}$:

$$\sum_{|\sigma|=k} \pi_\sigma^{1-\alpha} \delta_\sigma^{\alpha\beta(\alpha)} \sim 1. \quad (11.6)$$

Step 31. Comparing eq. (11.6) with the generalized energy function (eq. 5.2): setting $q = 1 - \alpha$ (so $q < 1$ for $\alpha > 0$) and $x = \alpha\beta(\alpha)/(q - 1) = \alpha\beta(\alpha)/(-\alpha) = -\beta(\alpha)$ —but since $l_\sigma = \delta_\sigma$ and the exponent in (5.2) is $(1 - q)x = \alpha x$, matching requires $\alpha\beta(\alpha) = (1 - q)D_q = \alpha D_{1-\alpha}$. Therefore:

$$\beta(\alpha) = D_{1-\alpha}(\mu). \quad (11.7)$$

This is the **fundamental Julia set relation**. It asserts that the quantum dynamical exponent at moment order α equals the Rényi dimension of order $1 - \alpha$. Since D_q is non-increasing in q , and $1 - \alpha$ decreases as α increases, $\beta(\alpha) = D_{1-\alpha}$ is non-decreasing in α —higher moments grow faster, reflecting the intermittent structure of the wave-packet.

3.14 General Linear IFS and Conjectures (Sections 12–13)

Step 32. For a general linear IFS with maps $l_i(s) = \delta_i s + \theta_i$ and probabilities π_i , the orthogonal polynomials satisfy a renormalization relation:

$$p_n(\mu; l_i(s)) = \sum_{k=0}^n \Gamma_{i,k}^n p_k(\mu; s), \quad (12.2)$$

where the coefficients $\Gamma_{i,k}^n$ are determined by the IFS parameters and the Jacobi coefficients of μ . These are identified as **Lanczos vectors** associated with a generalization of the Jacobi matrix. This replaces the exact polynomial renormalization (eq. 9.4) of the Julia set case, which had $\Gamma_{j,k}^{2n} = \delta_{nk}$ (only the $k = n$ term survives).

Step 33. For two-map linear IFS with $\pi_i = \delta_i^{D_0}$ (so that $\sum \delta_i^{D_0} = 1$, identifying D_0 as the similarity dimension = box-counting dimension by the open set condition), these are **uniform Gibbs measures**: $D_q(\mu) = D_0(\mu)$ for all q . This is because the measure assigns mass proportional to the D_0 -th power of the diameter to each cylinder set, making all local dimensions equal.

Conjecture 1: For two-map IFS with $\pi_i = \delta_i^{D_0}$, the intermittency function $\beta(\alpha)$ depends only on $D_0(\mu)$, not on the individual contraction ratios δ_1, δ_2 .

Step 34. Conjecture 1 fails for three or more maps. Even with $\pi_i = \delta_i^{D_0}$ and hence $D_q = D_0$ for all q , different three-map IFS with the same D_0 can have different $\beta(\alpha)$. The distinguishing feature is the **logarithmic equilibrium measure** of the support, which depends on the gap structure (the relative positions and sizes of the intervals in the Cantor set), not just on the similarity dimension.

Conjecture 2: For uniform Gibbs measures, if the equilibrium measures of two different supports have the same generalized dimension spectrum, then the intermittency functions coincide.

The restriction to Gibbs uniformity is essential: changing the weights π_i while keeping the contraction ratios δ_i fixed alters $\beta(\alpha)$ (since the generalized dimensions D_q become q -dependent), even though the equilibrium measure—determined solely by the geometry of the support—remains unchanged.

4. Clarification of Ambiguity

4.1 Notation for $J_n(\mu; t)$

The notation $J_n(\mu; t)$ is deliberately chosen to evoke the classical Bessel functions $J_n(t)$, which are recovered as a special case. However, the generalized F-B functions are not Bessel functions in any standard sense when μ is singular continuous. They do not satisfy Bessel's differential equation. The analogy is structural: $J_n(\mu; t)$ plays the role of $J_n(t)$ in the expansion of a plane wave e^{-its} in orthogonal polynomials, generalizing the Jacobi-Anger expansion $e^{-it \cos \theta} = \sum_n (-i)^n J_n(t) T_n(\cos \theta)$.

4.2 Meaning of $\beta^\pm(\alpha)$

The exponents β^+ and β^- correspond to **lim sup** and **lim inf** respectively (eq. 1.9). When $\beta^+ \neq \beta^-$, the growth of $\nu_\alpha(t)$ does not follow a single power law—there exist subsequences of times with different effective exponents. This is the mathematical content of "quantum intermittency." The fundamental relation (eq. 11.7) is stated for $\beta(\alpha)$ without the $\pm$ decoration; the numerical evidence in the paper suggests it holds for both β^+ and β^- in the Julia set case, but the renormalization argument strictly yields a relation for the quantity controlling the dominant scaling.

4.3 Convention $\omega = t^{-1}$

The inverse relation $\omega = t^{-1}$ means that statements about "small ω " are equivalent to statements about "large t ." The Gaussian kernel width ω in the spatial (spectral) domain corresponds to the time scale $t = \omega^{-1}$ in the dynamical domain. This is the standard uncertainty relation for Gaussian wave-packets. All asymptotic statements about $\omega \rightarrow 0$ are equivalently about $t \rightarrow \infty$.

4.4 The Index q in Generalized Dimensions

In eq. (5.1), the generalized dimension $D_q(\mu)$ is defined for $q \neq 1$. The case $q = 1$ requires a separate limit (L'Hôpital): $D_1(\mu) = \lim_{\omega \rightarrow 0} \frac{\int \log(\mu(B_\omega(r))) d\mu(r)}{\log \omega}$, which is the information (entropy) dimension. In the fundamental relation (eq. 11.7), the argument $1 - \alpha$ ranges over $(-\infty, 1)$ as α ranges over $(0, \infty)$. For $\alpha = 0$, one gets $\beta(0) = D_1(\mu)$, which is consistent with the interpretation that the zeroth moment growth rate reflects the entropy dimension. For $\alpha > 1$, the index $1 - \alpha < 0$, and the generalized dimensions D_q for $q < 0$ probe the most rarefied (lowest local dimension) parts of the measure.

4.5 Diagonal Approximation

In the renormalization (Step 23), dropping off-diagonal terms ($\sigma \neq \sigma'$) is justified when the Cantor set has gaps large enough that different cylinder sets $\varphi_\sigma(\text{supp}(\mu))$ and $\varphi_{\sigma'}(\text{supp}(\mu))$ are separated by a distance exceeding ω . For the Julia set with $\lambda > 2$, the gap size at level k is proportional to $\lambda^{-k/2} \cdot (\lambda - 2)$, which exceeds ω for $k \lesssim \log(1/\omega)/\log \sqrt{\lambda}$. The approximation becomes exact in the limit $\omega \rightarrow 0$ at any fixed k , but the error analysis for the joint limit $k \rightarrow \infty, \omega \rightarrow 0$ is non-trivial and is one source of the conjectural status of the results beyond the Julia set case.

5. Illumination of Assumptions

5.1 Determinacy of the Moment Problem

Role: Ensures that μ is the unique measure with the given moment sequence, so that the orthogonal polynomials $p_n(\mu; s)$ and hence the F-B functions are well-defined.

Where used: In the definition (eq. 1.1) and throughout, to guarantee that the spectral measure of J_μ with respect to e_0 is μ .

If weakened: If the moment problem is indeterminate, there exist multiple measures with the same orthogonal polynomials, and the Jacobi matrix has deficiency indices $(1, 1)$. The F-B functions are still defined by eq. (1.1), but different self-adjoint extensions of J_μ lead to different spectral measures and potentially different asymptotic behavior.

5.2 Compact Support of μ

Role: Guarantees (i) the moment problem is determinate; (ii) the Jacobi matrix J_μ is a bounded operator on $\ell^2(\mathbb{Z}_+)$; (iii) the super-exponential decay (eq. 1.2) holds.

Where used: Eq. (1.2) and all bounds that rely on J_μ being bounded (finite propagation speed, $\beta^\pm \leq 1$).

If weakened: For measures with unbounded support (e.g., the Gaussian measure with Hermite polynomials), J_μ is unbounded, the super-exponential decay (1.2) no longer holds in general, and the growth exponents can exceed 1.

5.3 Singular Continuity of μ

Role: The non-trivial asymptotics (quantum intermittency, algebraic decay of Cesàro averages, non-constant $\beta(\alpha)$) arise specifically when μ is singular continuous.

Where used: The absence of absolutely continuous component ensures $J_n(\mu; t)$ does not exhibit the Riemann–Lebesgue decay $J_n \rightarrow 0$; the absence of a pure point component ensures the wave-packet does not remain localized.

If weakened: For absolutely continuous μ , one obtains $\beta(\alpha) = 1$ for all α (ballistic transport). For pure point μ (as in the Anderson model with localization), $\beta(\alpha) = 0$ for all α (dynamical localization).

5.4 IFS Structure (Sections 9–12)

Role: The IFS structure allows the renormalization argument. The polynomial renormalization (eq. 9.4) for Julia sets, and its generalization (eq. 12.2) for linear IFS, are the key tools that convert the multi-scale structure of the Cantor set into functional equations for ν_α .

Where used: Eqs. (9.5)–(9.8), (10.1)–(10.4), (11.5)–(11.7).

If removed: Without IFS structure, the renormalization argument does not apply. The general lower bounds (eqs. 3.2, 6.5) still hold, as they rely only on dimensional properties of μ and growth bounds on orthogonal polynomials, not on self-similarity. The fundamental relation (eq. 11.7) is specific to the Julia set (and conjecturally to certain IFS classes).

5.5 Equal Weights $\pi_j = 1/2$ for Julia Sets

Role: The balanced measure on the Julia set assigns equal probability to each inverse branch. This ensures $D_q = D_0$ for $q \leq 1$ and simplifies the renormalization: the factor $1/2^{2k}$ in eq. (9.6) and the subsequent algebra leading to eq. (11.7).

Where used: Eqs. (9.5), (10.2), (11.6).

If changed: For non-balanced measures on the Julia set (obtained by assigning unequal weights to the two branches), the generalized dimensions become non-trivially q -dependent, and the fundamental relation (11.7) would need modification.

5.6 Diagonal Approximation and Linearization

Role: Dropping off-diagonal terms and linearizing the IFS maps are the two approximations in the renormalization argument.

Where used: Steps 23–24, leading to eq. (9.8).

If not made: The exact renormalization (eq. 9.6) involves a sum over all pairs (σ, σ') of multi-indices. Retaining the off-diagonal terms would couple different scales and prevent the simple product-form scaling ansatz.

5.7 Uniform Bound on Polynomial Growth (eq. 6.4)

Role: The bound $\sum_{n=0}^N p_n^2(\mu; r) \leq CN^\gamma$ for all $r \in \text{supp}(\mu)$ is the crucial input from the theory of orthogonal polynomials. The exponent γ controls the "kinetic" contribution to wave-packet spreading.

Where used: Eqs. (6.5), (6.6), and in the derivation of $\gamma = 1$ for Julia sets (eq. 10.4).

If γ is larger: A larger γ weakens the lower bound on β^- . The value $\gamma = 1$ means the Christoffel function $1 / \sum_{n=0}^N p_n^2(\mu; r)$ decays like $1/N$, which is the "generic" rate for measures satisfying Stahl-Totik regularity conditions.

6. Consequences

6.1 Immediate Corollaries of the Fundamental Relation

Corollary 1. Setting $\alpha \rightarrow 0^+$ in eq. (11.7):

$$\beta(0^+) = D_1(\mu),$$

recovering (a version of) Guarneri's inequality as an equality for Julia set measures.

Corollary 2. Setting $\alpha = 1$:

$$\beta(1) = D_0(\mu),$$

linking the growth of the first moment (the mean position of the wave-packet) to the box-counting dimension of the support of μ .

Corollary 3. For $\alpha \rightarrow \infty$:

$$\beta(\alpha) \rightarrow D_{-\infty}(\mu) = \lim_{q \rightarrow -\infty} D_q(\mu),$$

which equals the supremum of local dimensions of μ —the dimension of the "most spread-out" part of the measure.

6.2 Non-Constancy of $\beta(\alpha)$ (Quantum Intermittency)

Since $D_q(\mu)$ is non-increasing in q and strictly decreasing for generic singular continuous measures (measures that are not uniform Gibbs), $\beta(\alpha) = D_{1-\alpha}$ is strictly increasing in α . This means higher moments of position grow faster than lower moments, confirming quantum intermittency: the wave-packet spreads in a non-uniform fashion, with rare but large excursions dominating high moments.

For uniform Gibbs measures (D_q constant), $\beta(\alpha) = D_0$ for all α , and intermittency is absent.

6.3 Connection to the Multifractal Formalism

The relation $\beta(\alpha) = D_{1-\alpha}(\mu)$ places the quantum dynamics problem squarely within the multifractal formalism. The Rényi dimension spectrum D_q encodes the distribution of local scaling exponents of μ (via the Legendre transform connecting D_q to the singularity spectrum $f(\alpha_{\text{loc}})$). The quantum dynamical exponents inherit this multifractal structure.

6.4 Verification Without Time-Averaging

The paper reports that numerical evidence supports eq. (11.7) even without Gaussian time-averaging—i.e., for the instantaneous $\nu_\alpha(t)$, not just $A_G(\nu_\alpha)(t)$. If confirmed rigorously, this would strengthen the result by eliminating the need for the smoothing operation.

6.5 Potential-Theoretic Interpretation (Section 13)

The observation that equilibrium measures distinguish IFS with the same D_q spectrum but different gap structures suggests a deeper connection between quantum dynamics on Jacobi matrices and logarithmic potential theory. This is unexpected: the Jacobi matrix is defined by the orthogonality measure μ , but the equilibrium measure is a property of the support of μ alone (it does not depend on μ except through the set $\text{supp}(\mu)$). Yet $\beta(\alpha)$ depends on both—as shown by the failure of Conjecture 1 for three-map IFS and the additional constraint in Conjecture 2.

7. Examples and Illustrations

7.1 Classical Bessel Functions: $\lambda = 2$

At $\lambda = 2$, the Julia set is $[-2, 2]$, the balanced measure is the (rescaled) Chebyshev measure, and $J_n(\mu; t) = (-i)^n J_n(t)$. The generalized dimensions are $D_q = 1$ for all q (the measure is absolutely continuous with a density that, despite the inverse square-root singularities at ± 1 , has all $D_q = 1$). The fundamental relation gives $\beta(\alpha) = D_{1-\alpha} = 1$ for all α : ballistic transport. The position moments grow as $\nu_\alpha(t) \sim t^\alpha$, consistent with the classical asymptotics of Bessel functions: $J_n(t) \sim \sqrt{2/(\pi t)} \cos(t - n\pi/2 - \pi/4)$ for $t \gg n$, and the main contribution to $\sum n^\alpha |J_n(t)|^2$ comes from $n \sim t$.

7.2 Julia Set Cantor Measure: $\lambda > 2$

For λ slightly greater than 2 (e.g., $\lambda = 2.5$), the Julia set is a Cantor set of Hausdorff dimension $D_0 = \log 2 / \log(\sqrt{\lambda}) < 1$. The balanced measure is singular continuous. The F-B functions $J_n(\mu; t)$ no longer decay monotonically in t ; instead they exhibit irregular oscillations. After Gaussian averaging, $A_G(|J_n(\mu; t)|^2)$ decays algebraically as $t^{-D_2(\mu)}$. The wave-packet spreads sub-ballistically: $\beta(\alpha) = D_{1-\alpha} < 1$. For the first moment, $\beta(1) = D_0 < 1$, meaning the mean position grows as t^{D_0} .

7.3 Sparse Barrier Model (Section 7)

For the sparse barrier Jacobi matrix with $V_\eta(k) = k^\eta$ at sparse sites, the upper bound (eq. 7.2) gives, for $\alpha = 2$:

$$\beta^-(2) \leq \frac{3}{2\eta + 3}.$$

For $\eta = 1$ (linear barriers): $\beta^-(2) \leq 3/5 = 0.6$. Meanwhile $\beta^+(2) = 1$. The separation $0.6 < 1$ quantifies intermittency: on average the wave is slow, but occasionally it propagates ballistically.

7.4 Worked Trace: Renormalization at $k = 1$

Consider the Julia set IFS with maps $\varphi_\pm(s) = \pm\sqrt{s + \lambda}$, and take $k = 1$ renormalization step. At site $N = 2n$, eq. (9.5) gives:

$$\psi_{2n}^G(t) = \frac{1}{4} \left[\int \int \chi_\omega(\varphi_+(r) - \varphi_+(s)) p_n p_n d\mu d\mu + \int \int \chi_\omega(\varphi_-(r) - \varphi_-(s)) p_n p_n d\mu \right]$$

The diagonal terms have $\varphi_+(r) - \varphi_+(s) \approx \frac{r-s}{2\sqrt{r_0+\lambda}}$ (linearizing around a fixed point r_0), so $\chi_\omega(\varphi_+(r) - \varphi_+(s)) \approx \chi_{\omega/\delta_+}(r - s)$ with $\delta_+ = (2\sqrt{r_0 + \lambda})^{-1}$. Each diagonal term equals $\psi_n^G(t \delta_\pm)$. Summing the two diagonal terms and multiplying by 1/4:

$$\psi_{2n}^G(t) \approx \frac{1}{4} [\psi_n^G(t \delta_+) + \psi_n^G(t \delta_-)].$$

By symmetry of the Julia set, $\delta_+ = \delta_- =: \delta$, so:

$$\psi_{2n}^G(t) \approx \frac{1}{2} \psi_n^G(t \delta).$$

This is the $k = 1$ case of eq. (9.8). Iterating: $\psi_{2^k n}^G(t) \approx 2^{-k} \psi_n^G(t \delta^k)$.

7.5 Numerical Verification of $\beta(\alpha) = D_{1-\alpha}$

The paper reports that for the quadratic Julia set at specific values of $\lambda > 2$, direct numerical computation of $\nu_\alpha(t)$ for several values of α yields growth exponents $\beta(\alpha)$ that match $D_{1-\alpha}(\mu)$ computed independently from the scaling of $I_q(\mu; \omega)$. The numerical wave-front

exponent $\eta = 0.84165$ provides an independent check: $\beta^+(\alpha) \leq \eta$, and the computed $D_{1-\alpha}$ satisfies this constraint for all tested α .

8. Demythologization

8.1 F-B Functions Are Matrix Elements, Not Special Functions

Despite the name and notation, the F-B functions $J_n(\mu; t)$ are matrix elements of the unitary group generated by a tridiagonal matrix. The integral representation (eq. 1.1) is the spectral-theorem rewriting of $\langle e_n, e^{-iJ_\mu t} e_0 \rangle$. No special-function theory (Bessel differential equation, recurrence in t , asymptotic expansion via steepest descent) is available in the general case. The "Bessel" in the name serves only as an anchor to the one case where everything is explicitly computable.

8.2 Quantum Intermittency Is Non-Self-Averaging

The intermittency phenomenon— $\beta^+ \neq \beta^-$ —means that position moments at different times t are not well-described by a single power law. This is not a failure of the averaging procedure; it is a property of the dynamics. The wave-packet undergoes extended periods of slow spreading punctuated by rapid advances. The Cesàro or Gaussian average tames this behavior enough to extract algebraic scaling laws, but the non-self-averaging nature is intrinsic.

8.3 The Renormalization Is an Approximate Functional Equation

The renormalization equations (9.8), (10.1), (11.5) are not exact. They are asymptotic functional equations valid in the regime where:

- The diagonal approximation holds (gaps in the Cantor set exceed ω).
- The linearization of the IFS maps is accurate (the support of μ is small relative to the nonlinearity scale of φ_j).

The power-law scaling ansatz (e.g., $\nu_\alpha(t) \sim t^{\alpha\beta(\alpha)}$) is a fixed-point solution of these functional equations. The argument does not prove that the actual $\nu_\alpha(t)$ satisfies the exact scaling; it identifies the scaling exponent that is consistent with the renormalization structure. Rigorous validation requires controlling the error terms $E(k, n, t)$.

8.4 The Role of D_q for $q < 0$

The fundamental relation $\beta(\alpha) = D_{1-\alpha}$ for $\alpha > 1$ requires generalized dimensions D_q with $q < 0$. These dimensions probe the parts of the measure where $\mu(B_\omega(r))$ is smallest—the "thinnest" parts of the Cantor set. High moments $\nu_\alpha(t)$ with large α emphasize the most distant sites reached by the wave, which correspond precisely to the thinnest parts of the spectral measure (where the wave can propagate fastest because the Christoffel–Darboux kernel is largest). This explains the structural reason for the appearance of D_q with $q < 0$ in the quantum dynamics of high moments.

8.5 Conjectures Reflect an Incompleteness, Not a Mystery

Conjecture 1 (invariance of $\beta(\alpha)$ within two-map IFS equivalence classes) and Conjecture 2 (invariance under matching equilibrium measure spectra) are stated because the renormalization argument determines $\beta(\alpha)$ from the generalized dimensions of μ , but different IFS with the same $D_q(\mu)$ can have different dynamics. The missing ingredient is the precise role of the equilibrium measure (which encodes the geometry of the support independently of the probability weights). The conjectures are not appeals to mystery; they identify the boundary of what the current renormalization method can determine.

9. Final Statement

The paper defines Fourier–Bessel functions $J_n(\mu; t) = \int p_n(\mu; s) e^{-its} d\mu(s)$ as the Fourier transforms of orthogonal polynomials with respect to their orthogonality measure, and investigates their asymptotic behavior when μ is singular continuous.

Three asymptotic regimes are treated:

1. **Fixed order, large argument:** After Cesàro or Gaussian time-averaging, $A_G(|J_n(\mu; t)|^2)$ decays algebraically with an exponent governed by the correlation dimension $D_2(\mu)$.
2. **Fixed argument, large order:** $|J_n(\mu; t)|$ decays faster than any exponential in n , provided $\text{supp}(\mu)$ is bounded.
3. **Joint asymptotics (position moments):** The growth of $\nu_\alpha(t) = \sum n^\alpha |J_n(\mu; t)|^2$ is characterized by quantum dynamical exponents $\beta^\pm(\alpha)$. General lower bounds $\beta^-(\alpha) \geq D_{(1+\alpha)^{-1}}(\mu)$ (Guarneri-Schulz-Baldes-Tcheremchantsev) and $\beta^-(\alpha) \geq \frac{1}{\gamma} D_{1-\alpha/\gamma}(\mu)$ (using orthogonal polynomial growth of order γ) hold for arbitrary compactly supported measures. For Julia set measures of $z \mapsto z^2 - \lambda$ ($\lambda > 2$), a renormalization argument exploiting the polynomial identity $p_{2n}(\mu; \varphi_j(s)) = p_n(\mu; s)$ and the IFS balance equation yields the **fundamental Julia set relation**:

$$\beta(\alpha) = D_{1-\alpha}(\mu),$$

linking the quantum dynamical exponents to the Rényi generalized dimension spectrum of the spectral measure. Upper bounds $\beta^-(\alpha) \leq (\alpha + 1)/(2\eta + \alpha + 1)$ are established for sparse barrier Jacobi matrices.

Two conjectures extend the theory: Conjecture 1 asserts that for two-map IFS uniform Gibbs measures, $\beta(\alpha)$ is an invariant of the equivalence class labeled by $D_0(\mu)$. Conjecture 2 (stronger) asserts that for uniform Gibbs measures, the intermittency function is determined by the generalized dimension spectrum of the equilibrium measure of the support.

The essential hypotheses are: determinacy of the moment problem (guaranteed by compact support), the IFS self-similar structure of μ (for the renormalization argument), and the validity of the diagonal approximation and linearization (for the derivation of the functional equations whose scaling solutions yield the fundamental relation).

Source: Giorgio Mantica, "Fourier–Bessel functions of singular continuous measures and their many asymptotics," *ETNA* 25 (2006), 409–430; [arXiv:math-ph/0603069v1](#).